Spend Pixels Where It Hurts: Hazard-Aware Rendering for Optical Long-Context Compression

Supplementary Material

A. Implementation Details

Renderer. Pages are 1008 pixels wide (36 patches) with a 4-pixel margin and are rendered in grayscale with the Menlo monospace font through PIL. Prose is wrapped at character boundaries; a flagged word is placed whole on the next line when it does not fit. Each line is as tall as its largest glyph plus one pixel of leading. Documents taller than 2016 pixels are split into pages at blank pixel rows, so no line is cut. All pages are sent in one request in reading order. The budget of Eq. (1) is enforced on the unsplit image. Splitting can add one patch row per page, which made 10 of the 180 reallocated long-context images 1.2% more expensive than their uniform counterparts. All reported costs are billed tokens and include this effect.

Detector. After stripping the characters `.,;:()[]"'` from both ends, a word is flagged if it matches `[0-9a-f]{8,}`, `\d{5,}`, `/\S*\d\S*`, `\d+\.\d+\.\d+\.\d+`, `\S+/\S+\.\w{1,5}`, or `sha256=`. A word with at least one letter and one digit is also flagged if it contains `[A-Za-z0-9_\-]{20,}`. It is flagged as well if it is at least 16 characters long and has a 16-character window whose character entropy is at least 3.5 bits.

Baselines. U tries Menlo, Arial, and Arial Narrow at sizes 14 down to 5 px and keeps the largest size that fits the budget (ties broken by font order). At the 5 px and 6 px budgets it selects 6 px Arial and 8 px Arial Narrow for most documents. Z renders at 10 px and shrinks the image by repeated factors of 0.98 with Lanczos resampling until it fits the budget (final scale 0.52–0.64). For RULER and the natural-identifier questions, K and X append the line "Exact strings copied verbatim from the text block in the image:" after the question, followed by one flagged string per line. For LongBench, the anchors are inserted after the image placeholder and before the question, under "Exact strings copied verbatim from the text in the image:".

Prompts and scoring. RULER prompts follow the official template, with the context replaced by "[The text is rendered in the image above.]". They add "Copy each one exactly, character for character." and ask for a JSON list of values. A gold value counts as recalled if it appears (case-insensitively) in any returned string. LongBench prompts are the official ones for each task, and the first output line is scored with the official QA F1. For the plain-text arm the context is inserted verbatim into the same template.

Router threshold. τ is chosen per reader by two-fold cross-validation on the density sweep (folds by item parity). The criterion is recall minus 0.5 times the cost relative to plain text, maximized over $\tau \in \{0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.75, 1, 1.5, 2, 4, \infty\}$. Selected values were 0.1–0.5 with median 0.3, which we use for all other experiments. The frontier in Fig. 6c sweeps τ, and the per-density figures use the median τ. Both are therefore in-sample, unlike the cross-validated aggregate.

B. Per-Reader Results

Tables 5 and 16 give per-reader numbers for Figs. 3 and 6, and Tab. 6 gives the detector ablation of Sec. 5.7. In the length study, S is above A for every reader at every length. The size of the gain depends mostly on how well a reader does under uniform rendering. Claude and Gemini start at 0.45–0.57 and 0.21–0.47 under A, and S lifts them to 0.86–0.97 and 0.70–0.91. GPT, Qwen3-VL, Llama, and MiniMax recover at most 0.12 under A at every length, and S lifts them to 0.40–0.60. With length, S rises for Claude and MiniMax and falls for the other readers. Gemini falls most (0.91 to 0.70), and its uniform curve falls further (0.47 to 0.21). In the detector ablation, the regex and oracle detectors give the same recall when the needles are seven-digit numbers. With four-digit needles, which the regex detector deliberately ignores, the regex arms match uniform rendering while the oracle arms reach 0.82 (S) and 0.99 (X).

Table 5. Per-reader recall for the length study (Fig. 3). The mean excludes Claude.

	32K		64K		128K	
Reader	A	S	A	S	A	S
Claude Sonnet 5.5	0.57	**0.86**	0.51	**0.96**	0.45	**0.97**
Gemini 3.5 Flash-Lite	0.47	**0.91**	0.26	**0.74**	0.21	**0.70**
GPT-6 Luna	0.09	**0.59**	0.12	**0.55**	0.09	**0.50**
Qwen3-VL-235B	0.02	**0.60**	0.02	**0.55**	0.06	**0.54**
Llama 4 Maverick	0.01	**0.50**	0.00	**0.54**	0.00	**0.47**
MiniMax-M3	0.00	**0.40**	0.00	**0.49**	0.00	**0.56**
Mean	0.12	0.60	0.08	0.57	0.07	0.56

Table 6. **Detector ablation.** RULER multi-value recall (16K characters, 6 px budget, $N{=}25$, mean of 5 readers; Claude was not run in this ablation). The regex detector flags seven-digit values but deliberately skips four-digit ones, which collide with years. On four-digit values the regex arms therefore fall back to uniform rendering.

Arm	7-digit values (detected)	4-digit values (missed)
Uniform (A)	0.22	0.32
Ours, regex detector (S)	0.77	0.33
Ours, oracle detector	0.78	0.82
Ours + anchors, regex (X)	0.99	0.32
Ours + anchors, oracle	0.99	0.99

C. Statistical Procedures

Metrics. Recall is the fraction of gold values that appear, case-insensitively, as a substring of some returned value. The precision-aware F1 deduplicates the returned values, counts a returned value as correct when it contains a gold value, and combines this precision with recall. An empty answer scores zero. Exact-set accuracy requires every gold value to be found and no other value to be returned. Outputs that cannot be parsed are scored as empty answers. Empty responses and transport errors are dropped (Sec. 4).

Per-cell tests. For one reader on one task at one setting, we bootstrap the mean paired difference over items. The per-cell intervals in the tables use 10,000 resamples, and the p-values used for Holm correction use 1,000. A cell is significant when the two-sided bootstrap p-value is below 0.05.

Pooled effects. The pooled effect is the mean paired difference over all (run, reader, item) triples of a family, where a family is one row of Tab. 2. The item-level interval resamples items within each run, with the same item draw for every reader, so it treats the five readers as fixed. The reader-and-item interval also resamples readers with replacement. Because readers are crossed with runs, one reader draw is shared by all runs of the family. Both use 4,000 resamples and percentile intervals. Within each family we apply Holm's step-down correction to the per-cell p-values and report how many cells remain significant in each direction. Table 7 lists every comparison we computed, including those not discussed in the main text.

D. Precision-Aware Scoring and Anchor Content

For every RULER setting with anchors, Tab. 9 reports recall and precision-aware F1 for A, S, K, and X, and how many values each arm returns. It also reports two baselines that never look at the image: returning every anchor (dump) and returning as many random anchors as there are gold values (pick). Three observations qualify the anchor results in the main text. First, on single and multi-value the anchors are exactly the needles, so K and X match the dump baseline and their recall says nothing about reading. Second, on multi-key and in the density sweep at $k{\geq}4$, several readers (Qwen3-VL and MiniMax most often) return most or all of the anchor list. Recall then stays high while F1 falls to the level of the pick baseline. Third,

Table 7. **All pooled comparisons, RULER at 16K characters** (five non-Claude readers; recall and precision-aware F1). $\mathrm{CI}_{\mathrm{item}}$ resamples items within each run with the readers fixed. $\mathrm{CI}_{\mathrm{rd+item}}$ also resamples readers (one draw shared by all runs). Cells are counted as significantly positive/negative by per-cell bootstrap, raw and after Holm correction within the row.

Setting	Comparison	Metric	Δ	$\mathrm{CI}_{\mathrm{item}}$	$\mathrm{CI}_{\mathrm{rd+item}}$	raw +/−	Holm +/−	cells
RULER 16K, 5 px	S−A	recall	+0.55	[+0.50, +0.60]	[+0.48, +0.62]	14/0	14/0	15
		F1	+0.46	[+0.41, +0.50]	[+0.39, +0.54]	14/0	14/0	15
RULER 16K, 5 px	S−U	recall	+0.52	[+0.47, +0.56]	[+0.44, +0.60]	14/0	13/0	15
		F1	+0.42	[+0.38, +0.46]	[+0.37, +0.47]	13/0	13/0	15
RULER 16K, 5 px	S−Z	recall	+0.47	[+0.43, +0.52]	[+0.34, +0.59]	13/0	13/0	15
		F1	+0.38	[+0.34, +0.42]	[+0.29, +0.46]	13/0	13/0	15
RULER 16K, 5 px	S−R	recall	+0.54	[+0.49, +0.58]	[+0.47, +0.60]	14/0	14/0	15
		F1	+0.45	[+0.41, +0.48]	[+0.39, +0.50]	14/0	14/0	15
RULER 16K, 5 px	X−K	recall	+0.01	[-0.01, +0.02]	[-0.02, +0.04]	0/0	0/0	15
		F1	0.00	[-0.01, +0.02]	[-0.02, +0.03]	0/0	0/0	15
RULER 16K, 6 px	S−A	recall	+0.43	[+0.39, +0.47]	[+0.27, +0.56]	11/0	11/0	15
		F1	+0.34	[+0.30, +0.37]	[+0.21, +0.45]	11/0	9/0	15
RULER 16K, 6 px	S−U	recall	+0.44	[+0.40, +0.48]	[+0.31, +0.55]	11/0	11/0	15
		F1	+0.35	[+0.31, +0.38]	[+0.26, +0.43]	11/0	10/0	15
RULER 16K, 6 px	S−Z	recall	+0.47	[+0.42, +0.51]	[+0.30, +0.60]	11/0	11/0	15
		F1	+0.38	[+0.34, +0.41]	[+0.24, +0.49]	11/0	11/0	15
RULER 16K, 6 px	S−R	recall	+0.56	[+0.51, +0.60]	[+0.47, +0.64]	12/0	12/0	15
		F1	+0.47	[+0.43, +0.50]	[+0.38, +0.55]	12/0	12/0	15
RULER 16K, 6 px	X−K	recall	+0.05	[+0.03, +0.08]	[0.00, +0.12]	3/0	1/0	15
		F1	+0.04	[+0.01, +0.06]	[0.00, +0.08]	2/0	1/0	15
RULER 16K, 6 px (box runs)	B−A	recall	-0.05	[-0.07, -0.03]	[-0.13, 0.00]	0/3	0/1	15
		F1	-0.05	[-0.07, -0.03]	[-0.13, 0.00]	0/3	0/1	15
RULER 16K, 6 px (box runs)	S−B	recall	+0.53	[+0.50, +0.57]	[+0.41, +0.64]	14/0	14/0	15
		F1	+0.45	[+0.42, +0.49]	[+0.36, +0.56]	14/0	14/0	15
RULER 16K, 6 px (box runs)	S−A	recall	+0.49	[+0.45, +0.52]	[+0.29, +0.63]	13/0	13/0	15
		F1	+0.40	[+0.37, +0.44]	[+0.26, +0.53]	14/0	14/0	15

uniform rendering sometimes produces long lists of hallucinated identifiers (up to 55 per item for MiniMax at 5 px), which F1 penalizes and recall does not. The gains of S over A are therefore smaller under F1 but remain significant (Tab. 7). On the natural-identifier questions, readers return one value per question for every arm, so recall and F1 coincide.

Table 10 gives the rate of well-formed near-misses on RULER. A gold value counts as a near-miss when it was not recalled but some returned string has the same length and at least 60% character similarity. Allocation does not reduce this rate. On average it rises from 0.12 to 0.15 at 5 px and from 0.08 to 0.15 at 6 px. A plausible cause is that allocation turns many complete misses into nearly correct readings, some of which are still wrong. Anchors nearly remove near-misses where they contain the answer. We therefore do not claim that allocation makes errors detectable; check-and-retry [1] addresses that problem.

E. Controls, Milder Budgets, and Replication

Table 11 gives per-reader results for the box control and the milder budgets of Secs. 5.1 and 5.3, and Fig. 4 shows one item from the box runs. The box runs used a fresh sample of 25 items per task and re-read A and S, so they also replicate Tab. 1. On the new items, A averages 0.20 and S 0.68, against 0.22 and 0.65 on the original items at the same budget. At 8 and 10 px the U arm switches to Arial Narrow at 10 and 13 px, so it catches up with S at the lighter budget. Anchors (K, X) reach 0.96–1.00 at both budgets because the anchors again contain the needles.

F. Anchors: Budget Frontier and Long Context

Table 12 gives the per-reader numbers behind the frontier of Sec. 5.5: the 60 flagged natural-identifier questions rendered at base sizes of 5, 6, and 7 px. Allocation alone gains 0.19–0.25 over uniform rendering at every budget. Anchors add 0.28–0.37 on top of S and 0.46–0.53 on top of A. Table 13 gives multi-key results at 64K and 128K characters. Recall separates the arms but F1 does not, because every arm with enlarged or copied values tends to return several candidates for the one queried key.

Table 8. **All pooled comparisons, other settings** (five non-Claude readers). On natural strings, recall and F1 coincide. On LongBench, only F1 is defined. $\mathrm{CI}_{\mathrm{item}}$ resamples items within each run with the readers fixed. $\mathrm{CI}_{\mathrm{rd+item}}$ also resamples readers (one draw shared by all runs). Cells are counted as significantly positive/negative by per-cell bootstrap, raw and after Holm correction within the row.

Setting	Comparison	Metric	Δ	$\mathrm{CI}_{\mathrm{item}}$	$\mathrm{CI}_{\mathrm{rd+item}}$	raw $+/-$	Holm $+/-$	cells
RULER 32K–128K, 6 px	S−A	recall	+0.48	[+0.46, +0.51]	[+0.43, +0.53]	39/0	35/0	45
		F1	+0.38	[+0.35, +0.40]	[+0.34, +0.41]	35/0	30/0	45
natural, flagged, 5 px	X−K	recall	+0.04	[-0.01, +0.09]	[-0.03, +0.11]	0/0	0/0	5
natural, flagged, 5 px	X−K	F1	+0.03	[-0.02, +0.08]	[-0.03, +0.10]	0/0	0/0	5
natural, flagged, 6 px	S−A	recall	+0.25	[+0.18, +0.31]	[+0.12, +0.38]	4/0	4/0	5
natural, flagged, 6 px	S−A	F1	+0.25	[+0.18, +0.31]	[+0.12, +0.38]	4/0	4/0	5
natural, flagged, 6 px	X−A	recall	+0.52	[+0.45, +0.60]	[+0.33, +0.68]	5/0	5/0	5
natural, flagged, 6 px	X−A	F1	+0.52	[+0.44, +0.59]	[+0.33, +0.67]	5/0	5/0	5
natural, flagged, 6 px	X−K	recall	+0.02	[-0.04, +0.08]	[-0.06, +0.10]	0/0	0/0	5
natural, flagged, 6 px	X−K	F1	+0.02	[-0.04, +0.08]	[-0.06, +0.10]	0/0	0/0	5
natural, flagged, 6 px	K−A	recall	+0.50	[+0.42, +0.58]	[+0.31, +0.67]	5/0	5/0	5
natural, flagged, 6 px	K−A	F1	+0.50	[+0.42, +0.57]	[+0.30, +0.66]	5/0	5/0	5
natural, flagged, 7 px	X−K	recall	+0.05	[-0.01, +0.11]	[-0.06, +0.16]	1/0	1/0	5
natural, flagged, 7 px	X−K	F1	+0.05	[-0.01, +0.10]	[-0.06, +0.15]	1/0	1/0	5
natural, detector-indep.	S−A	recall	-0.01	[-0.07, +0.04]	[-0.11, +0.08]	0/0	0/0	5
natural, detector-indep.	S−A	F1	-0.01	[-0.07, +0.04]	[-0.11, +0.08]	0/0	0/0	5
natural, detector-indep.	X−A	recall	-0.09	[-0.16, -0.02]	[-0.19, 0.00]	0/2	0/1	5
natural, detector-indep.	X−A	F1	-0.09	[-0.16, -0.02]	[-0.19, 0.00]	0/2	0/1	5
natural, detector-indep.	X−K	recall	-0.08	[-0.13, -0.02]	[-0.17, +0.01]	0/1	0/1	5
natural, detector-indep.	X−K	F1	-0.08	[-0.13, -0.02]	[-0.17, +0.01]	0/1	0/1	5
natural, detector-indep.	K−A	recall	-0.01	[-0.07, +0.04]	[-0.08, +0.06]	0/0	0/0	5
natural, detector-indep.	K−A	F1	-0.01	[-0.07, +0.04]	[-0.08, +0.06]	0/0	0/0	5
natural, detector-indep.	X−T	recall	-0.59	[-0.66, -0.53]	[-0.75, -0.41]	0/5	0/5	5
natural, detector-indep.	X−T	F1	-0.59	[-0.66, -0.53]	[-0.75, -0.41]	0/5	0/5	5
density $k \leq 16$, 32K	S−A	recall	+0.32	[+0.28, +0.37]	[+0.15, +0.52]	11/0	9/0	15
		F1	+0.25	[+0.21, +0.29]	[+0.10, +0.43]	11/0	9/0	15
density $k \leq 16$, 32K	X−K	recall	+0.06	[+0.02, +0.09]	[-0.01, +0.13]	2/0	0/0	15
		F1	-0.01	[-0.04, +0.02]	[-0.07, +0.04]	2/2	1/0	15
RULER 32K, 8 px	S−A	recall	+0.38	[+0.32, +0.43]	[+0.28, +0.46]	8/0	8/0	10
		F1	+0.31	[+0.25, +0.36]	[+0.20, +0.42]	8/0	7/0	10
RULER 32K, 8 px	S−U	recall	+0.26	[+0.20, +0.34]	[+0.14, +0.40]	5/0	3/0	10
		F1	+0.23	[+0.17, +0.29]	[+0.10, +0.38]	5/0	3/0	10
RULER 32K, 8 px	X−K	recall	+0.03	[+0.01, +0.05]	[-0.01, +0.09]	1/0	0/0	10
		F1	0.00	[-0.02, +0.03]	[-0.04, +0.04]	0/0	0/0	10
RULER 32K, 10 px	S−A	recall	+0.20	[+0.14, +0.28]	[+0.11, +0.30]	4/0	4/0	10
		F1	+0.14	[+0.08, +0.21]	[+0.04, +0.25]	4/0	4/0	10
RULER 32K, 10 px	S−U	recall	+0.04	[-0.02, +0.10]	[-0.04, +0.12]	1/1	1/1	10
		F1	+0.03	[-0.03, +0.09]	[-0.05, +0.11]	1/1	1/1	10
RULER 32K, 10 px	X−K	recall	+0.03	[0.00, +0.06]	[-0.02, +0.08]	1/0	0/0	10
		F1	0.00	[-0.03, +0.03]	[-0.05, +0.06]	0/0	0/0	10
multi-key 64K–128K	S−A	recall	+0.66	[+0.59, +0.73]	[+0.50, +0.81]	10/0	10/0	10
		F1	+0.28	[+0.23, +0.33]	[+0.19, +0.36]	8/0	6/0	10
multi-key 64K–128K	X−K	recall	+0.03	[-0.01, +0.07]	[-0.04, +0.12]	1/0	0/0	10
		F1	-0.01	[-0.03, +0.02]	[-0.05, +0.03]	0/0	0/0	10
multi-key 64K–128K	K−A	recall	+0.84	[+0.80, +0.89]	[+0.63, +0.99]	10/0	10/0	10
		F1	+0.28	[+0.24, +0.31]	[+0.16, +0.35]	8/0	8/0	10
multi-key 64K–128K	X−A	recall	+0.87	[+0.82, +0.92]	[+0.73, +0.98]	10/0	10/0	10
		F1	+0.27	[+0.23, +0.31]	[+0.15, +0.36]	8/0	8/0	10
multi-key 64K–128K	X−S	recall	+0.21	[+0.14, +0.28]	[+0.09, +0.35]	5/0	3/0	10
		F1	-0.01	[-0.05, +0.03]	[-0.10, +0.08]	2/1	0/0	10
LongBench QA (F1)	S−A	F1	+0.02	[-0.01, +0.04]	[-0.02, +0.06]	0/1	0/0	10
LongBench QA (F1)	X−A	F1	+0.02	[-0.02, +0.05]	[-0.03, +0.07]	1/0	0/0	10
LongBench QA (F1)	K−A	F1	+0.04	[+0.01, +0.07]	[-0.01, +0.08]	2/0	0/0	10
LongBench QA (F1)	X−T	F1	-0.21	[-0.26, -0.16]	[-0.29, -0.13]	0/9	0/9	10

Table 9. **Precision-aware rescoring and anchor content** (means over five readers). rec: recall; F1: precision-aware F1 over the returned values; #ret: values returned per item; #anc: distinct anchors per item; dump: recall of returning every anchor; pick: expected recall of returning as many random anchors as there are gold values. Neither copy baseline looks at the image. On single and multi-value the anchor list contains exactly the needles, so K and X approach the dump baseline. At $k \geq 4$, several readers return most of the list, which recall rewards and F1 does not.

	A			S			K			X			anchor list		
Setting	rec	F1	#ret	rec	F1	#ret	rec	F1	#ret	rec	F1	#ret	#anc	dump	pick
16K, 5 px, single	0.02	0.02	1.4	0.42	0.41	0.9	0.99	0.99	1.0	0.98	0.98	1.0	1.0	1.00	1.00
16K, 5 px, multi-key	0.11	0.11	13.8	0.77	0.48	2.8	0.98	0.46	3.6	0.99	0.47	3.7	4.2	1.00	0.24
16K, 5 px, multi-value	0.12	0.13	10.1	0.73	0.76	3.5	0.96	0.95	4.0	0.97	0.97	4.0	4.2	1.00	0.96
16K, 6 px, single	0.16	0.16	1.1	0.40	0.40	1.1	0.98	0.98	1.0	1.00	1.00	1.0	1.0	1.00	0.98
16K, 6 px, multi-key	0.24	0.24	4.5	0.77	0.48	2.8	0.94	0.49	3.2	1.00	0.52	3.5	4.2	1.00	0.24
16K, 6 px, multi-value	0.25	0.28	3.5	0.79	0.81	3.6	0.90	0.91	3.7	0.99	0.97	4.1	4.3	1.00	0.96
density k=1	0.04	0.04	1.1	0.42	0.42	0.9	0.98	0.97	1.0	1.00	0.97	1.1	1.5	1.00	0.83
density k=4	0.03	0.03	1.1	0.40	0.25	2.9	0.93	0.48	3.2	0.98	0.42	3.8	4.0	1.00	0.25
density k=16	0.02	0.02	1.7	0.24	0.17	5.1	0.70	0.25	8.3	0.80	0.27	9.8	16.4	1.00	0.06
density k=64	0.03	0.03	2.8	0.06	0.06	5.6	0.34	0.16	10.0	0.32	0.13	10.8	64.2	1.00	0.02
multi-key 64K	0.11	0.11	4.8	0.79	0.39	3.3	0.94	0.38	4.2	0.99	0.40	4.3	4.8	1.00	0.22
multi-key 128K	0.11	0.10	16.6	0.75	0.38	3.3	0.96	0.39	4.7	0.97	0.36	5.1	6.0	1.00	0.18

Table 10. **Well-formed near-misses** on RULER (16K): fraction of gold values for which the reader returned a different string of the same length with at least 60% character similarity. Allocation does not reduce this rate; anchors nearly remove it.

	5 px budget					6 px budget				
Reader	A	U	Z	S	K	A	U	Z	S	K
Gemini	0.50	0.32	0.13	0.00	0.00	0.06	0.15	0.09	0.00	0.00
GPT	0.05	0.03	0.01	0.15	0.01	0.08	0.07	0.12	0.15	0.00
Qwen	0.01	0.02	0.01	0.25	0.00	0.14	0.12	0.14	0.14	0.00
Llama	0.02	0.02	0.01	0.25	0.00	0.05	0.17	0.10	0.21	0.02
MiniMax	0.00	0.00	0.00	0.13	0.00	0.07	0.07	0.08	0.23	0.00
Mean	0.12	0.08	0.03	0.15	0.00	0.08	0.11	0.10	0.15	0.00

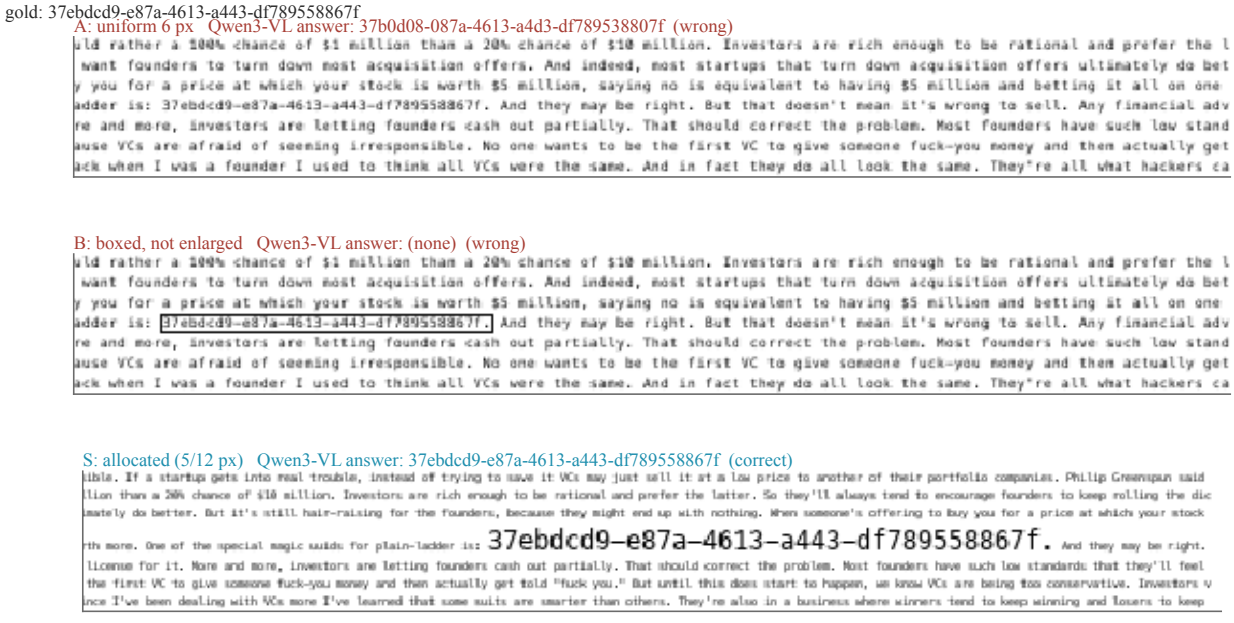


Figure 4. **One RULER item at the 6 px budget** (crops around the needle; Qwen3-VL answers). Uniform rendering yields a well-formed UUID with several wrong characters. Boxing the needle at the same size does not help (the reader returned nothing). Allocation draws the UUID at 12 px and prose at 5 px, and the answer is exact.

Table 11. **Salience control and milder budgets** (recall per reader). The box control averages three RULER tasks on a fresh sample of 25 items each; the milder budgets average single and multi-key. B boxes the flagged strings without enlarging them.

	Box control, 16K, 6 px			32K, 8 px					32K, 10 px				
Reader	A	B	S	A	U	S	K	X	A	U	S	K	X
Gemini	0.87	0.67	1.00	0.48	0.42	0.90	0.96	1.00	0.68	0.84	0.92	0.98	1.00
GPT	0.09	0.07	0.77	0.14	0.34	0.54	1.00	0.98	0.50	0.68	0.74	0.90	1.00
Qwen	0.02	0.00	0.58	0.14	0.28	0.48	1.00	1.00	0.44	0.58	0.62	1.00	1.00
Llama	0.00	0.00	0.56	0.00	0.18	0.44	0.88	1.00	0.16	0.36	0.38	0.92	0.96
MiniMax	0.00	0.00	0.50	0.00	0.10	0.28	1.00	1.00	0.22	0.36	0.36	1.00	0.98
Mean	0.20	0.15	0.68	0.15	0.26	0.53	0.97	1.00	0.40	0.56	0.60	0.96	0.99
Billed	929	923	843	1913	1855	1890	1954	1931	3061	2870	2826	3101	2866

Table 12. **Natural identifiers at three budgets** (flagged targets, N=60, recall). Billed tokens average the four readers that report usage. X is at or above K at every budget and costs the same as K at 5 px, where prose cannot shrink.

	5 px				6 px				7 px			
Reader	A	S	K	X	A	S	K	X	A	S	K	X
Gemini	0.62	0.75	0.92	0.90	0.72	0.80	0.87	0.90	0.77	0.85	0.92	0.85
GPT	0.17	0.47	0.80	0.83	0.25	0.55	0.85	0.82	0.35	0.55	0.87	0.88
Qwen	0.03	0.32	0.68	0.75	0.12	0.48	0.77	0.78	0.20	0.48	0.77	0.77
Llama	0.03	0.17	0.53	0.58	0.05	0.20	0.55	0.62	0.10	0.37	0.67	0.83
MiniMax	0.03	0.22	0.63	0.68	0.03	0.39	0.65	0.67	0.15	0.27	0.62	0.73
Mean	0.18	0.38	0.71	0.75	0.23	0.48	0.74	0.76	0.31	0.50	0.77	0.81
Billed	936	940	1014	1017	1181	1035	1267	1128	1267	1218	1345	1306

Table 13. **Anchors at long context** (RULER multi-key, 6 px, N=20; recall and precision-aware F1). Allocation makes X 29–31% cheaper than K, but under F1 the three arms S, K, and X are indistinguishable.

	64K characters								128K characters							
	A		S		K		X		A		S		K		X	
Reader	rec	F1	rec	F1	rec	F1	rec	F1	rec	F1	rec	F1	rec	F1	rec	F1
Gemini	0.35	0.35	0.90	0.54	0.80	0.42	1.00	0.40	0.40	0.40	0.95	0.63	0.85	0.50	1.00	0.47
GPT	0.15	0.15	0.65	0.38	0.90	0.38	0.95	0.46	0.10	0.10	0.60	0.40	0.95	0.44	0.95	0.36
Qwen	0.05	0.05	0.85	0.34	1.00	0.36	1.00	0.36	0.05	0.01	0.80	0.34	1.00	0.31	1.00	0.31
Llama	0.00	0.00	0.90	0.43	1.00	0.37	1.00	0.38	0.00	0.00	0.85	0.33	1.00	0.35	0.95	0.33
MiniMax	0.00	0.00	0.65	0.28	1.00	0.37	1.00	0.39	0.00	0.00	0.55	0.22	1.00	0.33	0.95	0.32
Mean	0.11	0.11	0.79	0.39	0.94	0.38	0.99	0.40	0.11	0.10	0.75	0.38	0.96	0.39	0.97	0.36
Billed	2627		1809		2669		1851		4850		3407		4901		3458	

G. Allocation Statistics

Table 14 lists the glyph sizes that Eq. (2) selects. At the 5 px budget, prose cannot shrink, so almost every document uses the -1 px leading fallback and the image fills the budget. At 6 px, prose shrinks to 5 px and flagged strings reach 12 px, leaving 29–32% of the budget unused on RULER. On natural documents and LongBench, a third to two thirds of the documents keep prose at 6 px because they contain few flagged strings. The images still use 9–16% fewer vision tokens than the budget. When flagged strings become dense (k=256), the enlarged size falls to 7 px.

H. Short Documents: Mechanism Study

Before the long-context experiments, we studied 1.6K-character documents that fit one small image. Each has six identifiers and three one-word prose facts (“The release codename is …”). We used both synthetic text and real package manifests whose lines contain file paths and 43-character base64 hashes. This setting isolates the allocation mechanism but is not a

Table 14. **Sizes chosen by the allocation** (prose/flagged glyph height in px, with the number of documents), vision tokens of S relative to the uniform budget, and the fraction of documents that needed the -1 px leading fallback.

Setting	docs	most frequent $s_{\mathrm{lo}}/s_{\mathrm{hi}}$	$v(S)/B$	fallback
16K, 5 px	25	5/12 (25)	0.93	0.96
16K, 6 px	25	5/12 (25)	0.71	0.00
32K, 6 px	20	5/12 (20)	0.70	0.00
128K, 6 px	20	5/12 (20)	0.68	0.00
32K, 8 px	25	8/12 (21), 7/12 (4)	0.96	0.00
32K, 10 px	25	9/12 (20), 10/12 (5)	0.82	0.00
density k=16	25	5/12 (25)	0.81	0.00
density k=256	25	5/7 (25)	0.94	0.00
natural, flagged	60	5/12 (37), 6/12 (22), 5/9 (1)	0.84	0.00
LongBench MFQA	50	6/12 (22), 5/12 (19), 5/9 (5)	0.91	0.00
LongBench HotpotQA	50	6/12 (32), 5/12 (18)	0.90	0.00

deployment target: for such short inputs an image is often not cheaper than the text.

Reallocation helps every reader; prose can pay for it. Table 15 reports word-wrapped reallocation against uniform rendering. Identifier exact-match improves significantly in all 24 reader–condition cells. Prose probes are unaffected on synthetic text except for Llama at 14 characters per vision token (-0.39). On real manifests, weaker readers lose up to 0.38 of prose accuracy, because whole manifest lines are enlarged and prose shrinks from 6 to 5 px. On long documents, where hazards occupy a small fraction of the text, this cost does not appear (Tab. 4).

Table 15. Short documents ($\approx$1.6K characters, one image, N=30). Difference between word-wrapped reallocation and uniform rendering in identifier exact-match (ids) and in accuracy on three prose probes (prose). Column headers give data and characters per vision token. */†: 95% CI above/below zero.

	synthetic, 21		synthetic, 14		manifests, 14 (a)		manifests, 14 (b)	
Reader	ids	prose	ids	prose	ids	prose	ids	prose
Claude Sonnet 5.5	+0.35*	+0.07	+0.19*	+0.06	+0.21*	-0.10	+0.13*	-0.08
Gemini 3.5 Flash-Lite	+0.33*	+0.01	+0.04*	-0.01	+0.23*	+0.00	+0.19*	-0.01
GPT-6 Luna	+0.21*	+0.06	+0.41*	+0.03	+0.17*	-0.16†	+0.18*	+0.03
Qwen3-VL-235B	+0.04*	+0.01	+0.44*	-0.08	+0.06*	-0.04	+0.08*	-0.00
Llama 4 Maverick	+0.05*	+0.00	+0.66*	-0.39†	+0.18*	-0.38†	+0.27*	-0.38†
MiniMax-M3	+0.28*	-0.03	+0.57*	-0.02	+0.17*	-0.21†	+0.21*	-0.14†

The gain depends on density and reader. Figure 5 sweeps four densities from 7.8 to 21.1 characters per vision token on synthetic documents. The gain follows an inverted U whose peak shifts with reader strength. The strongest readers gain most at the highest density (Claude $+0.32$ and Gemini $+0.39$ at 21 characters per vision token), where they begin to fail. Weaker readers gain most at moderate densities ($+0.48$ to $+0.78$ at 11–14).

Readers do not know when they misread. Allowing an "[unreadable]" answer did not consistently reduce silent errors. MiniMax-M3 used it for at most 6% of identifiers. Its rate of wrong-but-well-formed answers fell in one of four conditions (uniform, 21 characters per vision token: 0.40 to 0.31) and rose in the other three. This supports preventing errors at render time rather than relying on the reader's self-assessment.

Agent-based readers can leak across items. In a pilot, Claude readers accessed through an agent interface read several images in one session. Among their wrong prose answers, 50–100% were names that appeared in *other* images of the same session, versus 0–10% for MiniMax-M3 queried statelessly. All agent-based readings reported in this paper therefore use a Latin-square design in which each agent sees each item in only one form (Sec. M).

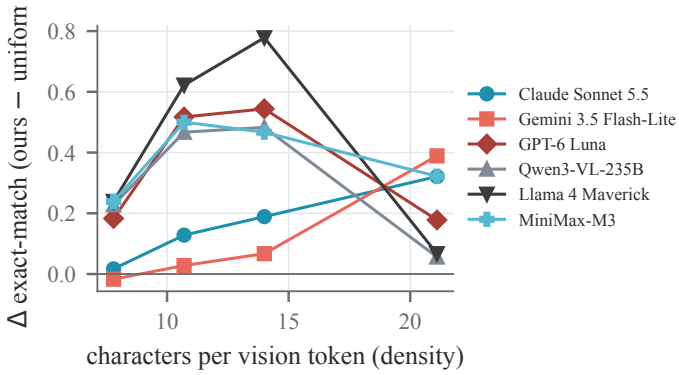


Figure 5. Gain in identifier exact-match of reallocation over uniform rendering versus rendering density (short synthetic documents, N=30 per point).

I. Negative Results

Global detector. An earlier detector, later replaced by the one in Sec. 3.2, also flagged four-digit numbers and long letter-only tokens. It flagged on average 85 and 112 words per MultiFieldQA and HotpotQA document, 62% of them years. On one LongBench sample it lowered F1 by 0.05 on average, with 3 of 12 cells significantly negative. On a second sample it was neutral (-0.005, 1 of 12 negative). With the final detector, the difference from uniform rendering is $+0.000$ (0 of 12 negative) on the second sample.

Word wrapping. Wrapping prose at word boundaries wastes part of every line. On long documents at the 5 px budget this left no room to enlarge anything, so that variant matched or exceeded the uniform cost while gaining little ($+0.00$ to $+0.12$ recall). Character wrapping for prose (Sec. 3.3) removes this waste.

Local instead of global shrinking. Shrinking only the prose within K words of a flagged string, with the smallest feasible K, was harmless on LongBench ($+0.003$, 0 of 12 negative). It retained only 84% of the RULER gain at the 6 px budget, because it fills the whole budget with slightly smaller enlarged strings. It brought no benefit over global shrinking.

Other variants. On short manifests, enlarging only the hash and path substrings of a manifest line freed almost no space. Adding one pixel of line spacing to prose-only lines did not change prose accuracy ($+0.00$, 95% CI $[-0.09, 0.09]$). Keeping prose at the uniform size left room only for 6–8 px identifiers and recovered 13–24 points less than full reallocation.

J. Hazard Density and Routing

Figure 6 varies the number of flagged strings per 32K-character document from 1 to 256 (per-reader values, including Claude, in Tab. 16). Allocation alone degrades as density grows (0.42 at k=1, 0.24 at k=16, 0.06 at k=64), because more strings compete for the pixels prose can give up. Anchors are strong while hazards are sparse but lose accuracy and grow expensive as hazards multiply. At k=64, K reaches 0.34 recall for 1.8× the tokens of uniform rendering. At k=4, X recovers 0.98 of the needles while costing 13% less than uniform rendering and 20% less than K. Under the precision-aware F1, however, X and K do not differ (pooled over $k \in \{1, 4, 16\}$: -0.01, CI $[-0.07, 0.04]$), and several readers return most of the anchor list at $k \geq 4$. For routing, τ was chosen per reader by two-fold cross-validation on this sweep. The selected values have median 0.3. Sweeping τ traces the recall–cost frontiers of panel (c), on which routing with X matches or dominates routing with K. At τ=0.3 it reaches 0.97 recall at 0.69× the plain-text cost versus 0.93 at 0.70×. These frontiers are computed on the same sweep used to select τ and should be read as in-sample.

K. Natural-Identifier Questions

For each of six LongBench tasks we shuffled the documents of at most 40K characters with a fixed seed and kept the first ten that contain a usable target. A target is a word flagged by the detector of Sec. 3.2. After surrounding punctuation is stripped, it must have 6–64 ASCII characters including a digit and start with a letter or digit. It must also be preceded by six words,

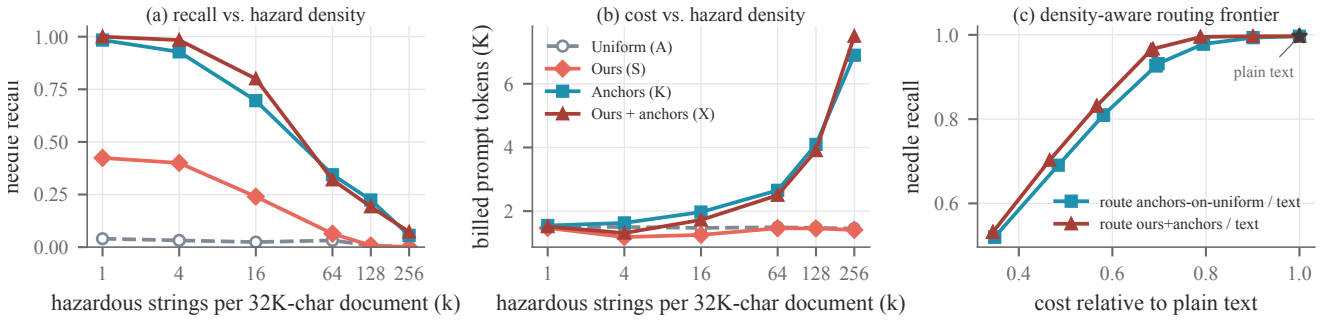


Figure 6. **Hazard density.** 32K-character documents with k UUID needles, one of which is queried (N=25). (a) Recall and (b) billed tokens for uniform rendering (A), our allocation (S), anchors on uniform rendering (K), and our allocation with anchors (X). Recall averages the five non-Claude readers; cost averages the readers that report usage (Claude included, MiniMax excluded). (c) Recall versus cost relative to plain text when Eq. (3) routes each document either to an image with anchors or to plain text. Panel (c) sweeps τ and uses the four readers with billed costs for both images and plain text.

Table 16. Per-reader recall in the hazard-density sweep. Each cell gives K/X (anchors on uniform rendering / on our rendering).

Reader	k=1	4	16	64	128	256
Claude Sonnet 5.5	1.00/1.00	1.00/1.00	0.88/0.96	0.60/0.72	0.28/0.44	0.16/0.24
Gemini 3.5 Flash-Lite	0.96/1.00	0.96/1.00	0.68/0.88	0.36/0.28	0.28/0.16	0.08/0.08
GPT-6 Luna	0.96/1.00	0.88/1.00	0.60/0.92	0.32/0.32	0.20/0.12	0.12/0.12
Qwen3-VL-235B	1.00/1.00	1.00/1.00	1.00/1.00	0.04/0.04	0.00/0.00	0.00/0.00
Llama 4 Maverick	1.00/1.00	0.80/0.96	0.48/0.44	0.48/0.20	0.16/0.16	0.08/0.04
MiniMax-M3	1.00/1.00	1.00/0.96	0.72/0.76	0.52/0.76	0.48/0.52	0.00/0.12

none of them flagged. The six-word phrase must occur exactly once in the document. We exclude dataset artifacts: Qasper reference placeholders (e.g., `BIBREF`), formula fragments, and characters replaced by "?" during ASCII normalization. We also exclude words glued across a missing space (a comma directly followed by a letter, a period or closing parenthesis followed by an uppercase letter, and a few letter–digit patterns). Targets containing "=" are also excluded. A first version of the set without the last two filters contained such artifacts in 14 of its 60 questions. Because anchors and enlarged rendering present the glued string as a single unit, that version favored our arms, and we discarded it. The prompt asks for the exact string that follows the phrase, and scoring uses the same recall as for RULER. The same 60 questions were also rendered at base sizes of 5 and 7 px for the frontier of Sec. 5.5.

Detector-independent set. A second set (new seed) measures what happens to strings the detector misses. It samples the target uniformly from all tokens of 4–64 ASCII characters that contain a digit. The phrase and artifact filters are the same (including `EQREF` and `DISPLAYFORM` placeholders), but the detector need not flag the target or the document. The detector flags one of the 60 targets.

L. Real-Corpus Profile

Table 17 lists the routing profile of real corpora. These are 13 English LongBench tasks (20 chunks each) and, with 36–40 chunks each, Paul Graham essays, Python source files from an installed package environment, local system logs, and Python package `RECORD` manifests. Every chunk is rendered and reallocated so that r uses the actual image cost.

M. Additional Claude Reader

We also used Claude, through an agent interface, as a reader that never informs the choice of any setting. For $k \in \{1, 4, 16, 64, 256\}$, ten items of the density sweep were read as X, K, and plain text in a Latin square. Each of 15 agents read ten different items, each in one form, using only a file-reading tool and a tool to write its answer file. Costs are estimated with Eq. (1) for images and with the linear text-token fit of Tab. 17. Table 18 shows that this stronger reader tolerates higher

Table 17. Routing profile of real corpora (32K-character chunks, τ=0.3). Flags per 1K characters, median and 90th-percentile routing ratio r, share routed to image+anchors (X), and estimated cost relative to plain text. Text tokens are estimated with a linear fit to billed text-only reads (0.208 tokens per ordinary and 0.737 per flagged character; mean absolute relative error 1.1%).

Corpus	n	flags/1K	r med	r p90	→X	cost/T
LB/narrativeqa	20	0.06	0.00	0.01	95%	0.27
LB/qasper	20	0.18	0.03	0.13	100%	0.19
LB/multifieldqa_en	20	0.85	0.05	0.98	75%	0.43
LB/hotpotqa	20	0.17	0.01	0.11	95%	0.25
LB/2wikimqa	20	0.06	0.01	0.05	100%	0.21
LB/musique	20	0.06	0.00	0.02	100%	0.22
LB/gov_report	20	0.04	0.00	0.02	100%	0.21
LB/qmsum	20	0.00	0.00	0.00	100%	0.23
LB/multi_news	20	0.06	0.00	0.05	100%	0.21
LB/triviaqa	20	0.23	0.08	0.19	95%	0.23
LB/passage_retrieval_en	20	0.04	0.00	0.02	100%	0.21
LB/lcc	20	1.40	0.40	0.83	25%	0.81
LB/repobench-p	20	1.43	0.52	0.87	5%	0.96
pg_essays	36	0.01	0.00	0.00	100%	0.22
python_source	40	2.62	0.67	1.47	12%	0.91
system_logs	40	11.84	3.73	4.09	0%	1.00
pkg_manifests	40	11.32	16.09	16.72	0%	1.00

density than the API readers (0.90 for X at k=64, against 0.60 for K). With τ=0.3, the routed system reaches 0.98 recall at an estimated 0.61× the plain-text cost. Agents may resample images when reading them, and N=10 per cell is small, so these numbers are indicative.

Table 18. Additional Claude reader (agent interface, one item per form per agent; N=10 per cell). Recall and estimated cost relative to plain text.

k	X	K	T	cost X/T	cost K/T
1	1.00	1.00	1.00	0.23	0.23
4	1.00	1.00	1.00	0.17	0.24
16	0.90	1.00	1.00	0.22	0.27
64	0.90	0.60	1.00	0.36	0.36
256	0.20	0.00	1.00	0.59	0.60

N. Reproducibility and Ethics

Prior work. The detector design, anchors, and text routing follow [1], which is under review; an anonymized copy is included in the supplementary material.

Code and data. The renderer, detector, router, task generators, reader clients, analysis scripts, all logged model outputs, and the scripts that produce every figure and table will be released. Task data are generated from public sources (RULER templates with Paul Graham essays, LongBench) with fixed seeds.

Ethics. The study involves no human subjects and no personal data. The system logs profiled in Sec. L were processed locally to count flagged strings; their contents are not released. Better verbatim fidelity under compression could in principle help extract identifiers from rendered documents. The method, however, only changes how a document the user already holds is presented to a model.